A Additional Experimental Details

A.1 Implementation Details

The Transformer denoiser has four blocks with hidden dimensions $[32, 32, 24, 24]$ and four attention heads. Training uses Adam with learning rate 10^{-3}, weight decay 5×10^{-4}, and dropout 0.1.

During training, the forward noising process uses $T = 1000$ steps. During inference, DDIM-based accelerated sampling uses a subsampled timestep schedule with $T_{\text{rev}} = 10$ denoising steps. These inference configurations use seed 0. OGB-molhiv and OGB-molpcba use $K = 5$ samples and an inference batch size of 1024 graph pairs; OGB-code2 uses $K = 5$ and a batch size of 512. IMDB uses $K = 10$, with inference batch sizes of 8192 and 256 for small and large graph pairs, respectively, defined by $\max(n_1, n_2) \leq 10$ and > 10. The sparse row/column budget is denoted by k_{sp}: the molecular runs use $k_{\text{sp}} = 16$ from block $\ell_{\text{sp}} = 1$ (zero-based), and IMDB uses $k_{\text{sp}} = 50$. Score calibration uses $\lambda = 1$ in $\mathbf{P}_t = \mathbf{S}_t - \lambda \mathbf{B}$.

Beam search uses beam width B, truncation factor Q, branch factor W, and search-step budget r, with an anchor ratio of zero. The settings (B, Q, W, r) are $(256, 1024, 32, 48)$ for OGB-molhiv and OGB-molpcba, $(1024, 4096, 64, 64)$ for OGB-code2, and $(1024, 4096, 32, 48)$ for IMDB.

A.2 Constrained Greedy Decoding

Given a calibrated score matrix $\mathbf{P} \in \mathbb{R}^{n_1 \times n_2}$, one decoder call constructs a binary matching while enforcing that every row and column is selected at most once. As illustrated below, it proceeds in three stages. First, it selects the highest-scoring column in every row, producing a row-wise top-1 candidate matrix. Second, when several rows select the same column, it retains only the candidate with the largest score in that column; exact-score ties are broken deterministically by row index. The surviving entries form a legal partial matching because no two selected pairs share a row or column. Let $\mathcal{U}_1$ and $\mathcal{U}_2$ denote the remaining unmatched row and column sets, respectively. The decoder repairs the residual region by selecting the global maximum among the currently feasible scores, adding the corresponding pair, and masking its row and column. It performs at most $R_{\text{rep}} = 6$ such global-greedy rounds, stopping earlier if either residual set is exhausted. If both sets remain nonempty after this budget, the decoder enumerates their indices in ascending order and, in one batched operation, pairs the first $m_{\text{free}} = \min(|\mathcal{U}_1|, |\mathcal{U}_2|)$ corresponding positions. The result is a legal injective matching that saturates the smaller graph.

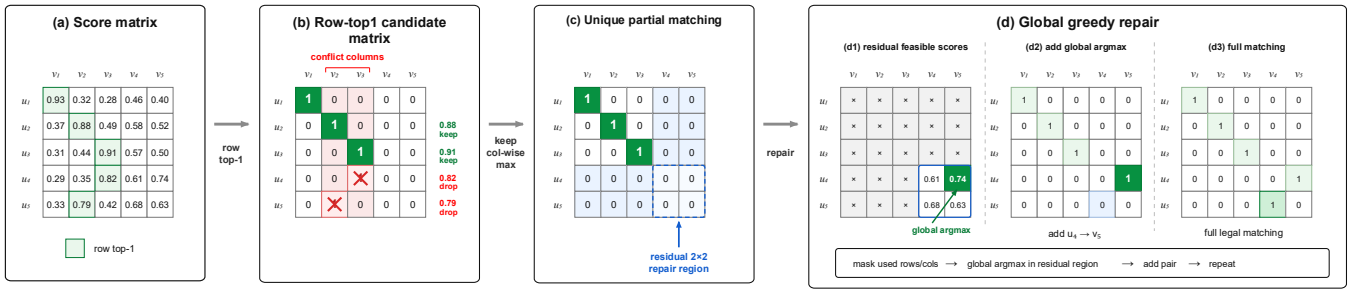


Figure 1: Constrained greedy decoding. Row-wise top-1 selection first proposes one candidate per source node. Column conflicts are resolved by retaining only the highest-scoring candidate in each conflicted column. The decoder then performs at most six global-greedy repair rounds in the residual feasible region; a deterministic batched fallback pairs corresponding residual nodes until the smaller side is exhausted.

Time and Space Complexity. The row-wise top-1 reduction scans the score matrix once and therefore requires $O(n_1 n_2)$ work. Conflict resolution processes the n_1 row candidates and aggregates them by column, requiring $O(n_1 + n_2)$ additional work. Let $q_{\text{rem}} = \min(|\mathcal{U}_1|, |\mathcal{U}_2|)$ be the number of pairs required to saturate the smaller residual side after conflict resolution. Each global-greedy repair round performs a masked argmax over all $n_1 n_2$ score entries and then masks one row and one column. With the fixed budget $R_{\text{rep}} = 6$, this phase performs only $\min(q_{\text{rem}}, R_{\text{rep}})$ sequential rounds and costs $O(\min(q_{\text{rem}}, R_{\text{rep}}) n_1 n_2)$. If $q_{\text{rem}} > R_{\text{rep}}$, constructing the remaining index lists and pairing their first $q_{\text{rem}} - R_{\text{rep}}$ positions costs $O(n_1 + n_2)$. A single decoder call therefore costs

$$O\big((1 + \min(q_{\text{rem}}, R_{\text{rep}}))n_1 n_2 + n_1 + n_2\big) = O(n_1 n_2),$$

because R_{rep} is a constant. With at most T_{rev} decoder calls per candidate and K candidates, the aggregate decoder work is bounded by $O\big(K T_{\text{rev}}(1 + R_{\text{rep}}) n_1 n_2\big)$. For fixed K, T_{rev}, and R_{rep}, the decoder work remains $O(n_1 n_2)$. The score, mask, and output matrices require $O(n_1 n_2)$ memory per candidate, or $O(b_{\text{pair}} K n_1 n_2)$ for a batch of b_{pair} graph pairs with K parallel candidates each.

Effect of the Repair Cap. Without a fixed cap, an $n \times n$ matrix can leave $q_{\text{rem}} = n - 1$ assignments after conflict resolution when all rows propose the same column. Completing these assignments with repeated dense global-maximum scans can require $\Theta(n^3)$ work. Fixing $R_{\text{rep}} = 6$ limits the number of such scans independently of graph size. The fallback enumerates the already ordered node indices by a linear scan, or by prefix-sum compaction on a GPU. Together, the fixed cap and linear fallback give an $O(n^2)$ worst-case decoder-work bound for arbitrary score matrices.

Proposition 1 (Exact Agreement Within the Budget). Given identical score matrices, initial partial matchings, and deterministic tie-breaking, bounded repair and unrestricted greedy completion return identical matchings whenever $q_{\text{rem}} \le R_{\text{rep}}$. If this condition holds for all K candidates of a graph pair, the minimum candidate GED is also identical.

Proof: Both procedures start from the same partial matching and apply the same greedy rule. By induction, their first $\min(q_{\text{rem}}, R_{\text{rep}})$ additions coincide. When $q_{\text{rem}} \le R_{\text{rep}}$, these additions saturate the smaller unmatched set within the repair budget, so fallback is never used. Applying this argument to every candidate preserves all K costs and therefore their minimum. □

Proposition 2 (A Sufficient Score-Separation Condition). Consider unmatched sets $\mathcal{U}_1, \mathcal{U}_2$ before a parallel proposal/conflict-resolution pass, with $m_{\text{free}} = \min(|\mathcal{U}_1|, |\mathcal{U}_2|)$. Suppose there are $m_{\text{free}} - s_{\text{miss}}$ rows $\mathcal{I}_{\text{sep}} \subseteq \mathcal{U}_1$ and an injective map $\varphi_{\text{inj}} : \mathcal{I}_{\text{sep}} \to \mathcal{U}_2$ such that, for some $\gamma > 0$,

$$\mathbf{P}[i, \varphi_{\text{inj}}(i)] \ge \mathbf{P}[i, j] + \gamma \qquad \text{for all } i \in \mathcal{I}_{\text{sep}}, \quad j \in \mathcal{U}_2 \setminus \{\varphi_{\text{inj}}(i)\}.$$

Then at most s_{miss} assignments remain after the pass. In particular, if $s_{\text{miss}} \le R_{\text{rep}}$, bounded and unrestricted repair agree without invoking fallback.

Proof: The rows in $\mathcal{I}_{\text{sep}}$ propose $m_{\text{free}} - s_{\text{miss}}$ distinct columns. Conflict resolution retains one assignment for every proposed column, even if another row outbids a row in $\mathcal{I}_{\text{sep}}$. Therefore at least $m_{\text{free}} - s_{\text{miss}}$ assignments survive, leaving at most s_{miss} to complete. Proposition 1 applies. □

Diffusion-Only Validation of the Six-Round Budget. This experiment tests whether six global-greedy repairs suffice to complete most final diffusion candidates without fallback, and how removing the cap affects GED estimates. On four datasets, we hold the final score matrices and partial matchings fixed and disable refinement, comparing six-round repair with uncapped greedy completion in terms of six-round coverage and mean GED difference. For each graph pair, the two decoders independently select their lowest-cost result from the same candidate set.

Table 1: Removing the default repair cap $R_{\text{rep}} = 6$, without refinement. Candidate coverage is the percentage of samples with $q_{\text{rem}} \le 6$; pair coverage requires all K samples to satisfy this condition. ΔGED is uncapped minus default Avg. GED.

Dataset	Candidate coverage (%)	Pair coverage (%)	ΔGED
OGB-molhiv	100.00	100.00	0.000000
OGB-molpcba	100.00	100.00	0.000000
OGB-code2	94.87	90.39	−0.000267
IMDB	96.50	93.49	−0.008933

Table 1 shows complete six-round coverage on molhiv and molpcba and high coverage on code2 and IMDB. On the latter two datasets, removing the cap leaves best-of-K GED unchanged for 99.83% and 99.82% of test pairs, respectively.

Ablating the Repair Cap during Search. We compare the default $R_{\text{rep}} = 6$ completion with an otherwise identical variant that removes the repair cap. Both variants use the same test pairs, frozen diffusion scores and initial matchings, and search settings. Fixed beam assignments and tie-breaking rules are preserved; only the default variant stops after six repairs and uses index-order fallback. Beam rankings, pruning, and subsequent trajectories evolve independently.

Table 2: Effect of removing the repair cap, relative to the default $R_{\text{rep}} = 6$. Fallback is the default's rate over active successor completions; changed pairs have different final costs. ΔGED is uncapped minus default Avg. GED; ΔTime is the relative change in search-runner time, excluding neural inference.

Dataset	Fallback (%)	Changed pairs (%)	ΔGED	ΔTime (%)
OGB-molhiv	0.449	0.49	−0.000067	+0.47
OGB-molpcba	0.318	0.13	−0.000333	+0.18

Table 2 shows that fallback is rarely needed during search. Removing the cap leaves final costs unchanged for nearly all graph pairs, with only a small change in average GED. The measured search times are similar for the two variants.